MIL-B-80012C
30 June 1988
SUPERSEDING
MIL-B-80012B
26 March 1971

MILITARY SPECIFICATION

BORING AND TURNING MACHINES, VERTICAL, AND LATHES, VERTICAL, TURRET, MANUAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers manually operated vertical boring and turning machines (vertical turret lathes) with equipment and accessories specified herein.

1.2 Classification. The machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2.1).

Size	Table diameter (inches)
3	36
4	46
5	56
6	66
7	72
8	84

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, DoD-adopted documents shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 70 - National Electrical Code.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B11.6 - Safety Requirements for the Construction, Care, and Use of Lathes.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

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(Application for copies should be addressed to the Joint Industrial Council, c/o National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22102-4269.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches.

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of precison boring and turning operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements of a manual, vertical, turret machine specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified (see 6.2.1), either US or SI units shall be used to graduate measuring and indicating devices. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position. When specified (see 6.2.1), indicating dials shall be graduated to read measurements in both US and SI Units.

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3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize its intended use, performance, or design life. The materials shall have been reclaimed in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI B11.6. If a conflict arises between 29 CFR 1910 and ANSI B11.6, 29 CFR 1910 shall apply.

3.2.5.1 Audible noise. Audible noise emitted by the equipment shall not exceed 84 dB at the operator's work position nor at any other point at a distance of 3 feet from the equipment, as measured on the "A" weight scale of a standard sound level meter under all operating and service conditions (see 4.7 and 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. Means shall be provided to indicate a low oil condition. All oil holes, grease fittings, and filler caps shall be accessible.

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3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.0001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.3.8 Gears. All gears, shafts, and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to

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meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of a base, column, rotating chuck or table, crossrail, turret or ram heads, side head, drive mechanisms, and operating controls.

3.4.1 Base and column. The base and column shall rigidly support the working components without appreciable deflection, which may be caused by the weight of the components and the workpart, and shall resist effects of cutting forces under maximum operating loads. The crossrail and side head shall be supported and guided by accurately finished guideways from a part of the column. The table drive mechanism shall be housed within the base. Means of leveling and securing the base to a foundation shall be provided.

3.4.2 Chuck. Unless otherwise specified (see 6.2.1), the machine shall be equipped with a four-jaw, independent chuck of the size specified in table I. The chuck shall serve as a table, shall rotate about a vertical axis, and be carried on not less than two large precision, preloaded, roller bearings, one radial and one thrust. The chuck furnished with the size 3 machine shall have not less than 8 radial T-slots. The chuck supplied with the size 4, 5, 6, 7, and 8 machines shall have not less than 12 radial T-slots. All T-slots shall conform to the applicable specifications of ANSI B5.1. Chuck jaws shall be arranged for internal and external grip on the workpart.

3.4.3 Chuck or table drive. Unless otherwise specified (see 6.2.1), the drive mechanism shall be infinitely variable through the speed range specified in table I. Chuck speed shall be adjustable under all cutting loads to obtain the most efficient cutting condition, within the selected speed range, without stopping rotation of the chuck. The speed transmission shall include rigid support for all gear shafts. The transmission housing shall fully protect the unit from damage by chips, dirt, and cutting fluids. Starting and stopping under load shall not impose undue stress on the drive mechanism.

3.4.4 Crossrail. The crossrail shall be heavily constructed, and shall provide guideways to support the rail heads under heavy cutting loads. The operation of one head shall not affect performance of a second head when operating simultaneously on finish cuts. Rail elevating screws shall be adjustable so that realignment of the rail on the column can be made. Adjustable gibs shall be provided to compensate for wear and permit adjustment between

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the rail and column. A clamping device shall be provided to lock the rail to the column. The clamping mechanism, if electrically operated, shall be safety interlocked with the crossrail elevating mechanism to prevent movement until the clamp is released. Upper and lower safety limit stops shall automatically disengage the rail elevating mechanism to prevent overtravel. Wipers shall be provided to protect all guideways from entrance of abrasive material between sliding components.

3.4.5 Rail head. Unless otherwise specified (see 6.2.1), the machine shall have a single rail head equipped with an indexing tool turret. The indexing mechanism shall include the machine builder's standard mechanism for indexing accuracy and clamping. When a single rail head is provided, it shall travel the full working length of the crossrail. When specified (see 6.2.1), except for the size 3 machine, table I, a ram type rail head shall be provided, in addition to the turret head, to operate on the left of the cross rail. Each rail head shall support maximum cutting loads of the machine throughout the slide travel. Each sliding unit shall have adjustable gibs to compensate for wear.

3.4.5.1 Taper turning. When required (see 6.2.1), taper turning capability shall be provided for the rail head, or heads, specified. The taper feature shall swivel not less than the amount specified in table I. The method of taper turning shall have an appropriate means for accurately selecting the required taper.

3.4.6 Side head. Unless otherwise specified (see 6.2.1), a side head shall be provided. The side head shall be mounted on the right-hand side of the machine, guided and supported by the right-hand vertical column way and a guideway on the side of the column. The side head shall contain the horizontal ram and feed mechanism. Unless otherwise specified, (see 6.2.1) the ram shall be equipped with a manually indexed, four-sided turret tool holder.

3.4.7 Feed transmission. Each rail head and the side head shall be provided with independent feed and rapid traverse in both horizontal and vertical directions of travel. The rate of feed shall be infinitely variable or selections shall be made in geometric progression. It shall be possible to adjust the rate of feed while the tool is cutting. Each head shall feed independently or simultaneously with another head. Feed movements of each head driven by feed screws shall be provided with means to adjust backlash (except recirculating ball screws). Suitable safety devices shall protect the feed mechanism from damage by overload and shall prevent engagement of feed and traverse controls simultaneously. Safety limits stops shall prevent overtravel of the feed units.

3.4.8 Counterbalancing. Rail heads and the side head, if provided, shall be counterbalanced to reduce wear and power requirement. When specified (see 6.2.1), vertical guideways or a similar means of preventing a pendulum motion of the counter weights shall be provided for shipboard installation.

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3.4.9 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.10 Hydraulic system. The hydraulic system shall meet JIC H-1 and be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.11 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit having a fused safety disconnect switch or circuit breaker and the machine shall be initially wired for operation on 460 volts.

3.4.11.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall have a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1 and Section 430-7 of ANSI/NFPA-70. Each motor shall have an identification plate to identify manufacturer, model, serial number, voltage, amperage, horsepower, phase, and frequency.

3.4.11.2 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.4.11.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input power supply by a low-voltage transformer having a secondary voltage of 120 volts or less.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than specified in table I for the machine size specified.

3.6 Performance. Each machine shall be capable of precision boring and turning operations as described in the tests specified in 4.7.

3.6.1 Additional performance requirements. Performance features required in addition to or in lieu of the requirements herein shall be as specified and fully described (see 6.2.1).

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3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II when tested according to 4.7.3.

3.8 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided. The light shall have a protective shield, and an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described shall not be required.

3.9 Standard equipment. All standard equipment required to properly operate the machine (such as special tools) shall be furnished.

3.9.1 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2.1). All repair parts shall be assigned a part number identified in a repair parts manual.

3.11 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Caution plate. Unless otherwise specified (see 6.2.1), a corrosion-resistant "CAUTION" or WARNING" plate shall be securely attached

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to the equipment in visible locations adjacent to danger zones. Safety precautions to be observied by the operator or maintenance personnel shall be permanently marked on these plates.

3.12 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.13 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.14 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.7).

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of temperature: 55°F to 85°F; and humidity: 10 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, any test, or the inspection shall be cause for rejection.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification, including compliance with all applicable standards cited in Section 2.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The machine shall be subjected to the following tests:

4.7.2.1 Horsepower Test. Each machine shall be subjected to a cutting test which applies not less than the full-rated input power to the spindle motor for at least one minute. Table speed shall be chosen by the supplier at the lowest practicable for this test without applying torque in excess of the design maximum value to any machine component. Material and dimensions of the workpiece, tooling, depth of cut, and feed rate shall be at the option of the supplier, provided the required loading is achieved. There shall be no apparent evidence of chatter, stalling, or permanent distortion during or after the test.

4.7.2.2 Turning test. A test part of SAE 1040 steel having a diameter of not less than 20 inches and a thickness suitable for a 12-inch turning cut shall be held by the chuck or on the table of the machine. Using a carbide turning tool, the outside diameter of the part shall be turned for a vertical length of not less than 12 inches. At least one such cut shall be made with

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each head. Upon completion of each cut, the part shall be checked for out-of-round accuracy using a three-point gage, checking at 0°, 120°, and 240°, and for turning cylindrical. All functions of the machine shall be operator initiated. Turning with rail heads and side head shall be out-of-round not more than 0.001 inch TIR; cylindrical taper per foot not more than 0.0015 inch when tested using the procedure given.

4.7.2.3 Audible noise test. Measurements shall be taken at the operator's work position and at each corner and at equidistant intervals of not more than 3 feet along each side and end of the equipment. For each measurement, the microphone shall be located on a straight line which is perpendicular to the surface/corner being measured and at a height corresponding to the point of highest noise level emitted from the surface/corner at herein specified location or distance from the equipment. Noise generated by the workpiece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

4.7.3 Accuracy test. Each machine and its components shall be tested for conformance with the accuracies specified in table II. The instruments and methods used in performing the tests shall be at the option of the supplier.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, (see 6.2.1), preservation, packaging, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The boring and turning machines and vertical turret lathes covered by this specification are for use in any machine shop where a part, due to its size, weight, or configuration, can be more conveniently handled on a horizontal table where the part rotates about a vertical axis. Operations include turning, boring, facing, and threading using modern cutting tools such as carbides.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system; dual unit dials, if required (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

i. Type chuck or table required, if different (see 3.4.2).

j. Type table drive, if different (see 3.4.3).

k. Single rail head, if different; ram type rail head when required for size 4, 5, 6, 7, or 8 machine (see 3.4.5).

l. Taper turning the rail head(s), if required (see 3.4.5.1).

m. Side head, if not required; tool holder; if not required (see 3.4.6).

n. Counterweight guideways, if required (see 3.4.8).

o. Electrical system, if different (see 3.4.11).

p. Motor enclosure, if different (see 3.4.11.1).

q. Size and capacity, if different (see 3.5).

r. Additional performance features, if required (see 3.6.1).

s. Worklight, if not required (see 3.8).

t. Specify and fully describe optional equipment, if required (see 3.9.1).

u. Specify applicable repair parts, if required (see 3.10).

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v. Lubrication plate, if different (see 3.11.1).

w. Nameplate, if different (see 3.11.2).

x. Caution plate, if required (see 3.11.3).

y. Hour meter, if required (see 3.12).

z. Inspection conditions, if different (see 4.3).

aa. First article inspection, if different (see 4.4).

bb. Quality conformance inspection, if different (see 4.5).

cc. If preservation, packing, and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operating manuals, parts, lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.4 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.5 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Inspection facilities. The contracting officer should be notified by each contractor in his response to the solicitation of the location where inspection tests are to be performed. It is recommended all inspections be performed within the limits of the continental United States.

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6.8 Subject term (keyword) listing.

Boring
Lathes
Manual
Operator
Rotating
Single
Turning
Turret
Variable
Vertical

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TABLE I. Sizes and capacities.

Characteristic	*Size 3	Size 4	Size 5	Size 6
Chuck or table diameter	36	46	56	66
Optional diameter	—	—	60	—
Swing, maximum diameter, with side head in working position	39	49	62	72
Swing, maximum diameter over side head	48	56	68	78
Height, maximum distance table to rail	30	40	48	60
Optional additional height under rail	12 or 24	12, 24, or 48	12, 24, or 48	12, 24, or 48
Rail travel, vertical	20	28	38	50
Speeds				
Range, infinitely variable, RPM	2.6 to 306	4.8 to 287	4.8 to 213	1.5 to 185
Feeds				
Number of selections, if not infinitely variable	20	20	20	20
Inches per revolution (IPR)	0.002 to 0.250	0.002 to 0.250	0.002 to 0.250	0.002 to 0.250
Optional high feed, IPR	0.500	0.500	0.500	0.500
Turret rail head				
Travel of turret slide, vertically	26	26	30	30
To center horizontally	43 (total)	30	38	43
**Swivel, right and left of vertical, degrees	30	30	30	30
Smallest bore turret tool enters, not more than	7-3/4	7-3/4	7-3/4	7-3/4
**Ram rail head				
Travel of ram, vertically	—	33	36	50
horizontally	—	32	38	90
**Swivel, right and left of vertical, degrees	< ——— 30 ——— >		< ——— 15 ——— >	
Smallest bore ram enters, not more than	11	11		
Side head				
Travel, vertically	28	38	46	58
Ram travel, horizontally	23	23	24	31
Motor horsepower, main drive	40	40	50	60
Optional horsepower	50,60,75	50,60,75	60,75,or100	75 or 100
Overall height, not to exceed dimension				
***specified by procuring activity				

All dimensions given in inches.

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

* One type rail head.
** When applicable.
*** See ordering data, 6.2.1(p).

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TABLE I. Sizes and capacities - Continued.

Characteristic	*Size 7	Size 8
Chuck or table diameter	72	84
Optional diameter	76	—
Swing, maximum diameter with side head in, working position	78	90
Swing, maximum diameter over side head	84	96
Height, maximum distance table to rail	70	70
Optional additional height under rail	12, 24 or 48	12, 24, or 48
Rail travel, vertical	50	50
Speeds		
Range, infinitely variable, RPM	3.8 to 200	3.2 to 160
Feeds		
Number of selections, if not infinitely variable	20	20
Inches per revolution (IPR)	0.002 to 0.250	0.002 to 0.250
Optional high feed, IPR	0.500	0.500
Turret rail head		
Travel of turret slide, vertically	30	30
To center horizontally	43	54
**Swivel, right and left of vertical	30°	30°
Smallest bore turret tool enters, not more than	7-3/4	7-3/4
**Ram rail head		
Travel of ram, vertically	48	48
horizontally	48	54
**Swivel, right and left of vertical, degrees	30	30
Smallest bore ram enters, not more than	15	15
Side head		
Travel, vertically	68	68
Ram travel, horizontally	31	31
Motor horsepower, main drive	50	50
Optional horsepower	75 or 100	75 or 100
Overall height, not to exceed dimension		
***Specified by procuring activity		

All dimensions given in inches.

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

* One type rail head.
** When applicable.
*** See ordering data, 6.2.1(p).

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TABLE II. Alignment tolerances.

Description of test	Tolerance (inches)	
Table surface, flatness, concave only	0.001	Per foot
Table face, true running	0.00025 TIR	Per foot radius
	0.003 TIR	Maximum
Vertical guideways, square with table face	0.0005	Per foot
	0.005	Maximum
Crossrail parallel to true plane of table surface	0.001	Per foot
	0.003	Maximum
Turret head slide and ram slide, side vertical travel parallel to table axis	0.0005	Per foot
Turret head free play, in locked position, measured at end of 8-inch test mandrel, 100 pound-feet torque	0.001	
Side head vertical travel parallel to table axis	0.0005	Per foot
	0.001	Maximum
Turret indexing, repetitive positioning within	0.0005	
Side head ram horizontal travel parallel to true plane of table surface, low to center	0.001	Per foot

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84

User activities:
Army - SM
Navy - YD

Preparing activity:
DLA - IP

Project (3411-0071)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-80012C	Boring and Turning Machines, Vertical, and Turret, Manual

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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